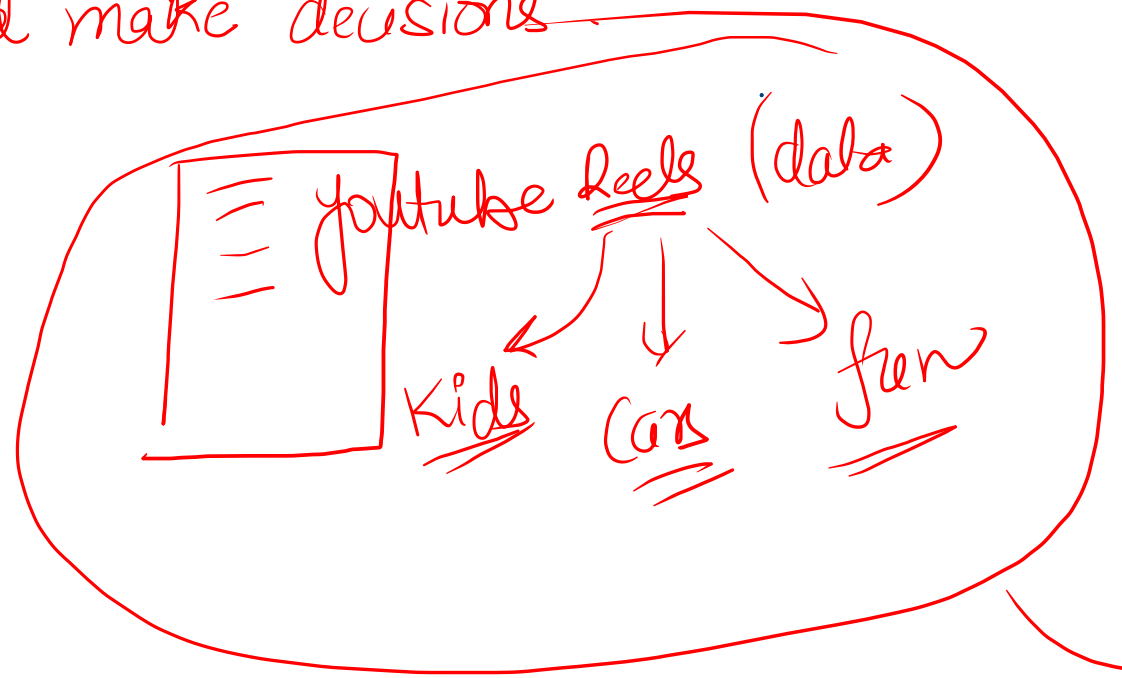
Data science

Good morning

Using data (information) to find answers and make decisions

collect data
↓
understand it
↓
use to predict or decide something



company

Fruitful Results

Wallmart

Kids
Clothing
Grocery
Electronics
Vehicles

10 million customers

Kids Clothing Grocery Electronics Vehicles

Maternity

10 million → ~~Kids~~

10 million → grocery

Electronics → ✓ 10 million

Choosing the Right customers is main part.

Datascience

Automated with AI

Amazon.com → You search for Shoes

Amazon will recommend similar Shoes

Recommends the products you like

Data science

Data Science actually does?

Flipkart (ecommerce)

└ collect data.

clean data (Python)

Analyze data

Models

Machine learning

manual job

① collect data

User clicks, purchases, search

② Clean data

Remove wrong or useless data.

③ Analyze data.

Find pattern like "People like X more ..

④ Make prediction

what you will like next

① Data collection

- Websites
- Apps
- Databases
- Sensors

} Python, SQL, ~~Apache~~ Apache Kafka

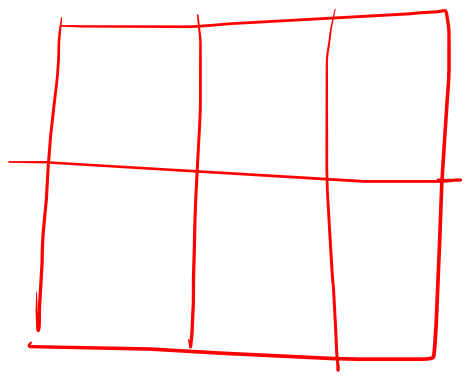
② Data cleaning

- Missing values
- Wrong entries
- Duplicate data

} Numpy (Python), Pandas

③ Data analysis → which product sells the most

→ when do users buy more?



Matplotlib

Sales increase on weekend

④ Data visualization → convert data into graphs & charts.

Tableau PowerBI

③ Machine learning (make predictions)

→ Smart way

→ Scikit learn

→ Tensor flow

(Netflix recommends that what you will watch next)

→ visual Studio code (VS code) — Install Them.

pip install ~~panas~~ pandas

→ Anaconda → A lot of python libraries → Pandas, Numpy

↓

Complete package for data science.

Python + libraries + Tools (all in one setup)

Python
Pandas
Numpy
Matplotlib
Jupyter Notebook

→ IDE (s/w where you write code)

→ very compatible to work with data science project.

Numpy is a library
used to work with numbers
fast and efficient using arrays

Numpy

Numerical python

CPU RAM

[1, 2, 10, 20, ...]

C++

C++

very fast

Numpy

[array]

Python API

AI

Python

Numpy

list → slow for calculation

It is the foundation of datascience
ML, AI, Pandas, ...

Numpy arrays faster than python list ✓

yes